VOLUME I · EXAM CHEATSHEET · STUDY GUIDE

CMS 706 — Database Systems

PGD Computer Science, Rivers State University, Nkpolu-Oroworukwo, Port Harcourt · built from the lecturer's 29 pages of class notes and the two past papers on file, one of them set by this same lecturer three weeks before this exam

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Exam: Friday 07 Aug 2026 **Time:** 11:00 – 14:00 **Venue:** the exam hall **Lecturer:** the lecturer

HOW THIS EXAMINER SETS A PAPER — EVIDENCE FROM THE PAST PAPERS

Two papers are on file. **DTS 304 Data Management I, dated 14/07/2026, was set by the lecturer** — the CMS 706 lecturer — and its content is almost exactly these notes. The older **CMS 445** paper is by different lecturers but covers the database half.

1. **Scenario questions, not bare definitions.** "A hospital replaces its paper records with a DBMS…", "A supermarket chain records millions of transactions daily — explain the difference between data and information *using this scenario*." Name the concept, then **apply it explicitly to the scenario given**. An answer that never mentions the hospital or the supermarket loses the application marks.
2. **SQL appears on every paper.** Write complete statements, uppercase keywords, ending in a semicolon.
3. **Both papers say "attempt any FIVE".** You get to leave one or two questions out — read everything first and choose deliberately.
4. **Lists earn marks per item.** Almost every concept in these notes came with a numbered list of advantages, disadvantages, benefits or challenges. Learn the lists, not just the definitions.

The lecturer's own paper of 14/07/2026, in full — seven questions: Q1 hospital DBMS hardware/software, benefits, implementation challenges · logical independence · data as a database component · user consent. **Q2** databases and data science · data masking · **data vs information** (supermarket) · **the data lifecycle** (healthcare, ten-year retention). **Q3** a data cleaning plan · **indexing** on 20 million records · GPS data for decision-making · **classifying data formats. Q4 design an ER diagram for a bank** (14 marks). **Q5** committee/professor queries · password policy. **Q6 the phases of query processing** · debugging two broken SQL queries. **Q7 the phases of data design** · concurrency control · define "transaction".

Read this carefully: questions 1, 2 and 3 are answerable almost entirely from these notes — **they are your five-question paper's safest three**. Questions 4 to 7 draw on material the class never reached: **ER diagrams, query-processing phases, data-design phases and division-style queries are all in Volume III**, and you need at least two of them to make up five answers. Learn this volume first, then Volume III §2–§6.

1 Definitions to write word-for-word

TERM	DEFINITION
Data	**Raw facts, figures or symbols** about people, things, events or processes that can be collected, measured and analysed. It is raw and unprocessed and **has no meaning on its own** — e.g. 25°C, or a person's age = 25.
Information	**Processed and organized data that is meaningful and useful** and helps in decision making — e.g. "Mary scored 25 marks in January." It is structured, organized and active.
Information management	The process of **collecting, storing, organizing and managing information efficiently**.
Data science	A **multi-disciplinary field that uses data, algorithms and scientific methods to extract insight and knowledge** from structured, semi-structured and unstructured datasets, combining computer science, statistics and machine learning to collect, clean, analyse and visualize data.
Data lifecycle	The **stages data goes through from creation to deletion**.
Metadata	**Data about data** — structured information that describes, explains, locates or otherwise makes it easier to retrieve, use or manage an information resource.
Information retrieval system	A system that **searches and retrieves relevant information from stored data** — e.g. search engines, library catalogue systems, e-commerce search.

TERM	DEFINITION
Indexing	A technique used to **speed up data retrieval by creating a reference (index) to data**, like the index in a book. Also: the process of creating a data structure that improves search speed.
DDDM	**Data-driven decision making** — the process of making decisions **based on data analysis rather than intuition**.
Information privacy	The **right of individuals or organizations to control how their data is collected, used and shared**.
Encryption	The process of **converting data into an unreadable format** — e.g. plain text "Hello" → "x592@H".
Data masking	**Hiding sensitive data** — e.g. showing a card number as `xxx 1234`.
Database	A **collection of interrelated data stored together with controlled redundancy to serve one or more applications in an optimal fashion**, organized so that a computer program can quickly select desired pieces of information. Also: an **integrated, self-describing collection of related data**.
DBMS	A **collection of related data and software programs used to define, construct, maintain and manipulate data in a database** — it enables you to store, modify and extract data. E.g. SQL Server, Access, Oracle.
Transaction	A **sequence of actions that represent a logical unit of work**, transforming the database from one state to another.
Database model	An **organizing principle that specifies a particular mechanism for data storage and retrieval**.
Database architecture	How the database is **structured** — how data flows between its parts; the **blueprint for storing, managing and accessing data efficiently**.
Logical data independence	The ability to **change the conceptual (logical) schema without changing the external schema or application program**.
Physical data independence	The ability to **change the physical storage scheme without changing the conceptual/logical schema**.

2 Data vs information PAST Q — SCENARIO

BASIS	DATA	INFORMATION
What it is	Raw, unprocessed facts, figures or symbols	Processed and organized data
Meaning	**Has no meaning on its own**	**Meaningful and useful**
Purpose	The raw material, awaiting meaning	Supports **decision making**
Form	Binary digital encoded (0, 1), measured in bits, bytes, gigabytes	Structured, organized and active
Dependency	Independent — exists on its own	**Depends on data** — cannot exist without it
Example	25, 28, 30 — bare marks	"Mary scored 25 marks in January"

Data by nature: quantitative — numeric information that can be measured (weight, height, sales figures, temperature) · **qualitative** — descriptive information expressing characteristics or attributes (customer feedback, colour of a car, satisfaction level).

ANSWERING THE SUPERMARKET SCENARIO

"A supermarket chain records millions of customer transactions daily. Explain the difference between data and information using this scenario."

The data is the raw transaction records themselves — every individual line: item purchased, price, quantity, till number, date and time. On their own these millions of rows mean nothing; no manager can look at them and decide anything.

The information is what emerges once those records are processed and organized — "bread sales rose 18% in Port Harcourt branches in June", "Saturday afternoon is the busiest trading period", "customers who buy nappies also buy milk". Each of these is **meaningful, useful and directly supports a decision**: what to restock, when to roster more staff, which products to shelve together.

The one-line summary: the supermarket *collects* data but *manages the business with* information — the transactions are the raw material, and processing them into sales trends is what turns them into information.

3 Information management and the data lifecycle PAST Q

CORE COMPONENTS OF INFORMATION MANAGEMENT (6)

1. **Data collection** — gathering raw data from sources (forms, sensors, logs)
2. **Data storage** — saving data in databases, cloud or files
3. **Data organization** — structuring data into tables, categories or format
4. **Data processing** — converting raw data into useful information
5. **Data retrieval** — accessing stored data when needed
6. **Data security** — protecting stored data from unauthorized access

OBJECTIVES OF INFORMATION MANAGEMENT (4)

Improved decision making · ensure data accuracy and consistency · enhance efficiency · support business operation.

REAL-WORLD EXAMPLES (3)

Banking systems managing transactions · hospital systems managing patient records · e-commerce platforms tracking orders.

THE EIGHT STAGES OF THE DATA LIFECYCLE

#	STAGE	WHAT HAPPENS
1	**Data creation**	Generation from sources — users or systems
2	**Data collection**	Gathering the data
3	**Data storage**	Saving it in databases or the cloud
4	**Data processing**	Cleaning and transforming
5	**Data analysis**	Finding patterns and trends
6	**Data distribution**	Sharing results with those who need them
7	**Data archiving**	Moving inactive data to long-term storage
8	**Data deletion**	Secure disposal at end of life

Importance of the data lifecycle: ensures data quality · improves data quality and usefulness · supports **compliance and security**.

Key insight to quote: *better data leads to better decisions.*

ANSWERING THE HEALTHCARE-RECORDS SCENARIO PAST Q · 8 MARKS

"*A healthcare company stores patient records for ten years before deletion. Outline and discuss the stages of the data lifecycle involved.*" — walk the eight stages **through that specific story**: **1. Creation** — the record is generated at the point of care: consultation notes, test results, vital signs. **2. Collection** — gathered into the hospital's system from wards, labs and monitoring devices. **3. Storage** — saved in the EHR database, with backups. **4. Processing** — cleaned, coded and transformed into a standard format so records from different departments agree. **5. Analysis** — used for diagnosis, treatment planning and disease prediction. **6. Distribution** — shared with authorized clinicians, and with insurers or regulators under consent. **7. Archiving** — after the patient becomes inactive, records move to long-term storage but must remain retrievable for the full **ten-year retention period**, which is a legal and compliance requirement, not a technical one. **8. Deletion** — at the end of the ten years the records are securely destroyed, because keeping personal health data longer than its purpose requires violates **data minimization** and privacy regulation such as HIPAA.

4 Data science, DDDM and data collection

HOW DATA IS USED IN DATA SCIENCE (5 STEPS)

1. **Data is collected** — from databases, APIs, sensors
2. **Data cleaning** — removing errors, duplicates, spaces
3. **Data analysis** — finding patterns and trends
4. **Model building** — predictive models, e.g. machine learning
5. **Decision making** — business insights and strategies

Examples: Netflix recommending movies · TikTok · banks detecting fraud · hospitals predicting disease.
Importance of data: data is the **foundation** of data science — **without data, analysis is impossible**.

DATA-DRIVEN DECISION MAKING — 6 STEPS

Data collection → data processing → data analysis → interpretation → decision making → prediction.
Example: sales data shows declining product demand → decision: reduce production or improve marketing.

ADVANTAGES OF DDDM	CHALLENGES OF DDDM
More accurate decisions	Poor data quality
Reduced risks	Lack of skills
Better performance	Data overload

THE THREE DATA COLLECTION TECHNIQUES

TECHNIQUE	WHAT IT IS · EXAMPLES	ADVANTAGES	DISADVANTAGES
Manual	Collected by humans without automation — surveys, questionnaires, paper forms, interviews, observation	Low cost and simple · flexible	Time consuming · prone to human error · difficult to scale
Sensor-based	Collected automatically by devices — thermometers, temperature sensors, biometric scanners, IoT, GPS	Real-time data · high accuracy · automation	Expensive · requires maintenance · may produce large data volumes
API-based	Collected from external systems via **Application Program Interfaces** — weather API, social media API, payment gateways	Fast and automated · access to large datasets · real-time updates	Requires programming knowledge · API limits/restrictions · depends on external systems

USE CASES OF DATA SCIENCE, BY SECTOR

Healthcare — patient records, disease prediction · **Banking** — fraud detection, transaction tracking · **E-commerce** — product recommendation, customer behaviour analysis · **Education** — student performance prediction, learning analytics · **Transportation** — traffic prediction, ride sharing.

Data formats — the classification question PAST Q

FORMAT	DEFINITION	CHARACTERISTICS	EXAMPLES
Structured	Data organized in a **fixed format**, usually in tables	Rows and columns · easy to store and query · highly organized	Student records · **SQL tables and a customer database** · Excel spreadsheets
Semi-structured	Data that **does not follow strict tables but has some structure**	Flexible structure · uses tags or key-value pairs	**JSON files · XML documents · emails**
Unstructured	Data with **no predefined format or structure**	Difficult to analyse · large in volume · requires advanced tools (AI, NLP)	**Videos · images · comments · audio recordings** · social media posts

The past-paper classification, answered. "*A social media platform stores videos, images and comments, XML documents, audio recordings, emails, a customer database and a JSON file — classify each.*"
Unstructured: videos · images · comments · audio recordings. **Semi-structured:** XML documents · emails · the JSON file. **Structured:** the customer database.
A JSON example to quote: `{ "Name": "John", "Age": 20 }` — it has keys and values, so it has *some* structure, but no fixed table.

5 Storage, retrieval, indexing and search

THE TWO STORAGE METHODS COMPARED

BASIS	FILE-BASED STORAGE	DATABASE SYSTEM (DBMS)
What it is	Data stored in **separate files** — text files, spreadsheets	A **structured system** where data is stored in tables and managed by a DBMS
Characteristics	Flat files (.txt, .csv) · each file independent · **no central control**	**Centralized** storage · tables of rows and columns · supports relationships (keys)
Advantages	Simple to use · low cost · suitable for small systems	Reduces redundancy · better data integrity · improved security · easy data retrieval · **multi-user access**
Disadvantages	**Data redundancy** · data inconsistency · difficult data sharing · poor security · **no relationship between data**	Very expensive · requires an expert or administrator · complex to manage

INFORMATION RETRIEVAL SYSTEM

Components (5): data collection · indexing system · query processor · search engine · user interface.
Retrieval process (4): user enters query → system searches indexed data → matches result → returns relevant information.
Types of retrieval: exact match (database query) · **best match** (search engines).

THE FOUR TYPES OF INDEX

INDEX	BASED ON
Primary index	The **primary key** — unique values
Secondary index	**Non-key attributes** — may have duplicates
Clustered index	Data **stored in order**
Non-clustered index	A **separate structure pointing to** the data

Benefits of indexing: faster search · improved query performance.
Limitations of indexing: takes extra storage · **slows down data updates** (every insert or update must maintain the index too).

THE 20-MILLION-RECORD BANK SCENARIO PAST Q

Without an index, a query must perform a **linear search** — checking all 20 million records one after another until the target is found, which is slow and grows worse as the bank adds customers. An **index** is a separate reference structure, typically a **B-tree**, holding the indexed values in sorted order with pointers to the actual rows. The query engine **searches the index, locates the pointer and retrieves only the matching record**, so it never scans the table. The **WHERE** condition is what benefits, and the effect grows with table size — which is exactly why it matters at 20 million rows and not at 20. The price is **extra storage** and **slower inserts and updates**, because every write must also update the index.

The two search techniques and the two index structures

TECHNIQUE	HOW IT WORKS	ADVANTAGES	DISADVANTAGES
Linear (sequential) search	All elements are checked one after another until the target is found: start at the first record, compare each value, stop when found or when the end is reached	Simple to implement · works on **unsorted** data	Slow for large datasets · poor time complexity

STRUCTURE	HOW IT WORKS	ADVANTAGES	DISADVANTAGES
B-tree indexing	A **balanced tree structure** storing **sorted** data. Nodes contain multiple keys; **all leaf nodes are at the same level**. Search by traversing tree levels	Efficient for large datasets · balanced structure gives **consistent performance** · **supports range queries**	Complex structure · requires maintenance

TECHNIQUE	HOW IT WORKS	ADVANTAGES	DISADVANTAGES
Index search	Uses an index to locate data **without scanning all records**: search the index, locate the pointer, retrieve the actual data	Much faster · efficient for large data	Requires extra storage · index maintenance overhead

Worked example from class: search for 25 in `{10, 15, 25, 30}` — linear search checks 10, then 15, then 25 and stops. An index search would go straight to it.

STRUCTURE	HOW IT WORKS	ADVANTAGES	DISADVANTAGES
Hashing	Uses a **hash function to map a key to a specific location** in memory — `Hash(25) → Address 5`	**Very fast look-up** · good time complexity	**Collision** — two keys map to the same location · **not suitable for range queries**

```
B-tree:      [20]
            /    \
        [10]   [30, 40]      all leaves on one level
```

6 Querying data efficiently and SQL

THE CLASS TABLE — `STUDENTDB`

S/N	NAME	MATNO	AGE	DEPT	SEX
1	John	2025/PGD/001	22	Agric Sc	M
2	Ada	2025/PGD/002	24	Economics	F
3	**Rueben**	2025/PGD/003	**30**	Maths	M
4	**Eze**	2025/PGD/004	**32**	Computer Sc	M
5	Obuchi	2025/PGD/005	21	Law	F

```
SELECT Name FROM StudentDB WHERE Age >= 30;

Result:  Rueben
         Eze
```

SIX TECHNIQUES FOR EFFICIENT QUERIES

1. **Use an index** — speeds up the `WHERE` condition
2. **Avoid `SELECT *`** — retrieve only the needed columns
3. **Use `WHERE` properly** — filter the data at the source
4. **Use `JOIN` efficiently** — join Table 1 `(Name, MatNo, Age, Dept, Sex)` to Table 2 `(State, LGA)` to get the combined row
5. **Limit results** — return only the rows you need
6. **Use aggregation carefully** — e.g. `SELECT COUNT(*) FROM Table1`

Query optimization concept: the **query optimizer chooses the best execution plan** — by reducing disk access, by reducing CPU usage, and by using indexes and statistics.

THE FOUR SUB-LANGUAGES OF SQL LEARN THE TABLE

CATEGORY	FULL NAME	WHAT YOU DO	KEYWORDS
DQL	Data Query Language	**Read** data	SELECT
DML	Data Manipulation Language	**Change** data	INSERT, UPDATE, DELETE
DDL	Data Definition Language	**Define** structure	CREATE, ALTER, DROP
DCL / TCL	Data Control / Transaction Control Language	**Permission and transaction**	GRANT, REVOKE, COMMIT, ROLLBACK

THE CORE STATEMENTS, AS WRITTEN IN CLASS

```
SELECT Name, Price
FROM Products
WHERE Category = 'iphones'
ORDER BY Price DESC
LIMIT 10;

INSERT INTO Users (Name, Email)
VALUES ('Ada', 'ada@gmail.com');

UPDATE Products
SET Price = Price * 1.1
WHERE Category = 'iphone';
```

Clause order never changes: `SELECT … FROM … WHERE … GROUP BY … HAVING … ORDER BY … LIMIT`. Writing `WHERE` after `ORDER BY` is a syntax error and an easy lost mark.

Real-world applications: e-commerce — fast product search using indices · banking — quick transaction retrieval · social media — searching users.

7 Privacy, integrity and security PAST Q

THE FIVE KEY PRINCIPLES OF PRIVACY

1. **Consent** — data should be collected with user permission
2. **Purpose limitation** — data used only for its intended purpose
3. **Data minimization** — collect only necessary data
4. **Transparency** — inform users how data is used
5. **Accountability** — organizations are responsible for data protection

FIVE DATA PROTECTION TECHNIQUES

TECHNIQUE	WHAT IT DOES
Encryption	Converts data into an unreadable format — "Hello" → "x592@H". Two types: **symmetric** and **asymmetric**
Access control	Restricts who can view or modify data — **role-based access control** and **user authentication** (password / biometrics)
Backup and recovery	Regular backups prevent data loss; recovery systems restore data after failure
Data masking	**Hiding sensitive data** — e.g. a card shown as `xxx 1234`

WHY USER CONSENT MATTERS PAST Q

Consent is the **first principle of information privacy** — the right of individuals to **control how their data is collected, used and shared**. Without it, collection is **unauthorized access** and exposes the customer to **identity theft and data misuse**. It is also a **legal requirement** under GDPR, HIPAA and Nigeria's data protection laws, so collecting without it risks penalties. And it is commercial: consent builds the **trust and transparency** that keep customers willing to share data at all. Consent also bounds the other principles — it is what defines the "intended purpose" that **purpose limitation** then restricts you to.

Types of data: personal (name, age) · financial (bank details) · health records · biometric · academic.
Privacy risks: identity theft · data misuse · unauthorized access.
Regulations: GDPR (Europe) · **HIPAA** (healthcare) · **local data protection laws** (Nigeria).

TECHNIQUE	WHAT IT DOES
Firewalls and antivirus	Protect against unauthorized access and malware

THREATS AND THE CONTROLS THAT ANSWER THEM

THREAT	CONTROL
Malware — viruses, worms, ransomware	**Firewalls** — block unauthorized network access
Phishing — fake emails to steal information	**Authentication** — verify user identity
Unauthorized access — hackers accessing systems	**Authorization** — define what users can access
SQL injection — malicious SQL commands	**Update and patching** — fix system vulnerabilities
Data breaches — exposure of sensitive data	**Encryption** · **monitoring and auditing** — track system activities

How data masking protects sensitive information: it **replaces or hides part of a sensitive value** while leaving the record usable — a card number displayed as `xxx 1234` still identifies the card to its owner but is **useless to anyone who sees it**, whether a call-centre agent, a developer testing on real data, or an attacker who obtains the screen. Unlike encryption it is not reversed for the viewer — the hidden digits are simply never shown.

8 Databases, the DBMS and its services

THE FOUR COMPONENTS OF A DATABASE

COMPONENT	MEANING
1. Data	Any computer representation of a stored logical entity — discrete pieces of information usually formatted in a special way
2. Relationship	Represents a **correspondence between various elements**
3. Constraints	**Predicates that define the correct state** of the database
4. Schema	Describes the **organization of and relationships between** the database — the design of the database

WHY SEVERAL TABLES? — THE CLASS EXPLANATION

Given Table 1 `(Age, Name, MatNo)`, Table 2 `(LGA, State)`, Table 3 `(Religion, Country)`, Table 4 `(PA, Sex, Height)` — the schema separates the data into tables that are **related because common attributes exist in a selected pair of the tables.** Because of those common attributes we can **combine two or more tables to get the complete data of each student**. That is exactly what a `JOIN` does.

THE DBMS

The DBMS is made up of **(a) the database, (b) the DBMS itself, (c) the application program** — what the user interacts with. Its **primary goal is to provide a mechanism where data can be stored and information retrieved** from the database.

```
  User
   ↓
┌──────────────────────────────┐
│ Application Process          │
│ ┌─────────────────────────┐  │      ┌──────────┐
│ │ Transaction Processing  │──┼─────▶│ DB       │
│ │ Data Management         │  │      │ metadata │
│ └─────────────────────────┘  │      └──────────┘
└──────────────────────────────┘ DBS
```

THE SEVEN SERVICES OF THE DBMS

SERVICE	WHAT IT DOES
1. Transaction management	A **transaction is a sequence of actions representing a logical unit of work**, transforming the database from one state to another. On **`COMMIT`** the changes are made **permanent**; on **`ROLLBACK`** the transaction returns to its original state.
2. Concurrency control	Coordinates **database manipulation processes that operate concurrently**, access shared data and can potentially interfere with one another. Its goal is to **allow concurrency while maintaining consistency** of the shared data.
3. Recovery management	Ensures that an **aborted or failed transaction does not adversely affect** the database or other transactions — the database is **returned to a consistent state** after a failure.
4. Security management	The **protection of data against unauthorized access** — only authorized users or administrators may reach the database.
5. Language interface	Provides the languages for the **definition and manipulation** of data — structures created with the **DDL**, manipulation done with **DML** commands.
6. Storage management	Manages the **permanent storage** of data. The **internal schema** defines how data is stored, and the storage manager **interfaces with the operating system** to reach physical storage.
7. Data catalog management	The **data catalog** is a system database holding **metadata** — information about data, relationships, constraints and the entire schema — organized into a unified database that **can itself be queried** to learn the structure of the database.

Why the DBMS replaced the file-based system

The **file-based system was developed in response to industry's need for more efficient data access**, but it has **three named limitations** — this is a standard exam question:

1. **Segregation and isolation of data** — when data is isolated in separate files it is more difficult to access data that should be available; the difficulty is compounded when data is required from more than two files.
2. **Duplication of data** — because each department takes a **decentralized** approach, the FBS encourages **uncontrolled duplication**, and duplication is wasteful.
3. **Data dependency** — the **physical structure and storage** of files and records are **defined in the application code**, so changes to an existing structure are difficult to make.

Applications of a database (7): purchasing from the supermarket · using credit cards · booking a vacation with a travel agent · a computerized library system · using the internet · taking out insurance · renting a video.

> **Note the symmetry worth writing:** each FBS limitation is answered by a DBMS advantage — **isolation** is answered by **centralized storage and relationships**, **duplication** by **controlled redundancy**, and **data dependency** by **data independence** (§10). Structuring the answer that way turns a list into an argument.

9 Database models

A **database model is an organizing principle that specifies a particular mechanism for data storage and retrieval**. The primary difference between models lies in **the methods of expressing relationships and constraints among the data elements**. The **relational model is the current favoured model**, while object-oriented and structured models are emerging.

#	MODEL	HOW IT ORGANIZES DATA	ADVANTAGES	DISADVANTAGES
1	**Hierarchical**	Organizes data elements as **tabular rows, one per instance of an entity**, in a **tree**: parent → child. E.g. a company's GM above departmental managers	Simple · has data security and integrity · efficient (not complicated) · fast access	Implementation complexity · database management problems · **lack of structural independence** · programming complexity · implementation limitation · **rigid, difficult to modify**
2	**Network**	**Replaces the hierarchical tree with a graph**, allowing more general connections among nodes. Its defining difference is the **many-to-many (n-n)** relationship — records may have **more than one parent**	Conceptual simplicity · handles more relationship types · ease of data access · data integrity and independence · database standards	System complexity · absence of structural independence
3	**Relational**	**Stores data in tables.** Powerful because it requires **few assumptions** about how data is related or extracted, so the **same database can be viewed in different ways** and a single database can be **spread across several tables**. Each table corresponds to an entity, each row to an instance	Simple and widely used · easy querying with SQL · flexible views	Less flexible for very complex data
4	**Object oriented**	Data stored as **objects**, as in programming	Handles complex data	Complex implementation
5	**Semi-structured**	A flexible model using formats like **JSON and XML**	Adaptable · used in modern web systems	Weaker guarantees than a fixed schema

Also named in the notes: the **object-relational model** and the **deductive model**.

THE BOOK / DISTRIBUTOR RELATIONSHIP EXAMPLE

```
Book ——— 1:1 ——— Distributor
   given a book, its distributor is determined

Book ——— n:1 ——— Distributor
   what books does this distributor supply?

Book ——— n:n ——— Distributor
   combines both; readable from either side
```

The notes use this to make one point: **the same real-world relationship can be depicted in several ways, and the model you choose decides which questions are cheap to answer.**

THE HIERARCHICAL EXAMPLE FROM CLASS

```
                 General Manager
      ┌────────────────┬────────────┬─────────────┐
   Sales       Treasury       Supply       Accounts
  Manager       Manager       Manager       Manager
      └────────────────┴────────────┴─────────────┘
                    Departments
```

Note the sentence the notes give for why the hierarchical model fell out of use: **many of its limitations result from its overly restrictive view of relationships** — a child may have only one parent, which real data often violates. That is precisely the gap the **network model's n-n relationship** was invented to fill.

10 Database architecture and independence

THE ANSI/SPARC THREE-LEVEL ARCHITECTURE

Used in most RDBMS — Postgres, MySQL, Oracle. **Database architecture is how the database is structured — how data flows between its parts; the blueprint for storing, managing and accessing data efficiently.**

THE TWO INDEPENDENCES — AND WHY THE

LEVEL	WHAT IT HOLDS
1. External / View	The **area or angle the users see** — views, reports, app interfaces. The user **cannot see the whole database**; complexity is hidden and security is added
2. Conceptual / Logical	**DBMS independent**; where the **DBA works**. It holds the "**what**" — tables, relationships, constraints and schemas
3. Internal / Physical	**How data is stored** — files, indices, data blocks, partitions, on disk or SSD

Types of database architecture: client–server · three-tier.

```
EXTERNAL   views · reports      ← what users see
CONCEPTUAL tables · relations   ← what the DBA designs
INTERNAL   files · indices      ← how it is stored
```

LEVELS EXIST

INDEPENDENCE	DEFINITION	CLASS EXAMPLE
Logical data independence	**The ability to change the conceptual (logical) schema without changing the external schema or application program**	You add a new column `Middle-Name` to the `Customers` table. If the application, running `SELECT First-Name, Middle-Name FROM Customers_Table`, keeps working, the database is **logically independent**
Physical data independence	**The ability to change the physical storage scheme without changing the conceptual/logical schema**	You move `Customers_Table` from **HDD to SSD**. If the application keeps working, you have **physical independence**

"HOW DOES ADDING NEW DATABASE FIELDS AFFECT LOGICAL INDEPENDENCE?" PAST Q

Adding a field changes the **conceptual/logical schema**. Logical data independence is precisely the property that this change **does not force a change to the external schema or the application programs** — existing views and queries that never mention the new column continue to run unchanged, because each external view exposes only the subset of the schema it was defined over.

So a well-designed three-level architecture **absorbs** the addition: the DBA adds `Middle-Name` at the conceptual level, and programs using `SELECT First-Name FROM Customers_Table` are unaffected. Independence **fails** only where an application depends on the schema's shape rather than on named columns — the classic case being `SELECT *`, which silently starts returning an extra column, or code that reads fields by position. **That is the strongest single argument for avoiding `SELECT *` in application code**, and it ties this question straight back to the efficient-query techniques in §6.

11 Sector applications — the table to memorise

SECTOR	DATA COLLECTED	APPLICATIONS USED	BENEFITS	CHALLENGES
Healthcare	Patient records (name, age, history) · medical images (X-ray, MRI) · lab results · vital signs (heart rate, blood pressure)	**EHR** — centralized patient data storage · **CDSS** — clinical decision support helping doctors diagnose · telemedicine — remote consultation · health monitoring systems — wearables	Improved patient care · faster diagnosis · reduced errors	Data privacy concerns · high system cost · integration issues
Finance	Transaction records · customer profiles · credit history · market data (stock monitoring)	Online banking systems · fraud detection systems · risk management systems · automated trading systems	Secure transactions · real-time processing · better financial decisions	Cyber security, threats and regulatory compliance

SECTOR	DATA COLLECTED	APPLICATIONS USED	BENEFITS	CHALLENGES
Logistics	Shipment details · inventory levels · delivery routes · GPS tracking	Supply chain management · inventory management · route optimization · warehouse management	Efficient delivery · reduced cost · real-time tracking	Data integration · system complexity
Social media	User profiles (text, likes, comments) · behavioural data · multimedia content	Recommendation systems · targeted advertising · content moderation · trend analysis	Personalized user experience · increased engagement · business insight	Privacy issues · misinformation · data overload

THE RIDE-SHARING GPS SCENARIO PAST Q

"*A ride-sharing company collects GPS locations every second. Explain how this data supports business decision-making.*" — this is a **DDDM question wearing a logistics costume**. Walk the six DDDM steps through it: the GPS stream is **collected** by sensor-based collection, **processed** and cleaned, **analysed** for patterns, **interpreted**, then used for **decisions** and **prediction**. Concretely, the company gains: **route optimization** — the shortest actual route given live traffic, cutting fuel cost and trip time · **demand forecasting** — which areas generate rides at which hours, so drivers are positioned before demand appears · **dynamic pricing** — surge where demand exceeds supply · **driver performance and safety monitoring** — speeding, harsh braking, idle time · **accurate ETAs**, which drive customer satisfaction and retention · and **expansion planning** — which new areas are underserved. Close with the DDDM definition itself: every one of these is a decision made **from data analysis rather than intuition**.

12 Things you can lose easy marks on

1. Defining data and information in the same words. **Data has no meaning on its own; information is processed, meaningful and supports a decision.**
2. Answering a scenario question with pure theory. **Name the concept, then apply it to the hospital, supermarket or bank you were given.**
3. Calling email or XML unstructured. They are **semi-structured** — they have tags or key–value pairs.
4. Saying indexing has no cost. It **takes extra storage and slows down updates.**
5. Claiming hashing is good for range queries. It is **not** — that is B-tree's advantage.
6. Confusing the two independences. **Logical** = change the logical schema, applications survive; **physical** = change the storage, the logical schema survives.
7. Mixing up the DBMS services. **Concurrency** control coordinates simultaneous transactions; **recovery** management handles failed ones.
8. Saying `COMMIT` can be undone. **`COMMIT` makes changes permanent; `ROLLBACK` is what returns the transaction to its original state.**
9. Putting `DELETE` under DDL. `DELETE` removes **rows** and is **DML**; `DROP` removes the **table** and is DDL.
10. Writing SQL clauses out of order or without a semicolon.
11. Giving the hierarchical model's n-n relationship. Many-to-many is the **network** model — hierarchical allows only one parent.
12. Listing three points where the question is worth 8 marks. **Marks are per point** — write the whole list.

TIMING PLAN FOR A 3-HOUR PAPER

Both past papers say "**attempt any FIVE**", so the first five minutes are spent **reading all the questions and choosing** — pick the five whose lists you know cold, and be willing to skip the one heaviest on unfamiliar material. Budget roughly **30 minutes per question**. For a scenario question, spend the first minute writing the **concept name and its definition**, then the rest applying it — that guarantees the definition marks before you risk anything on the application. For a question with parts (a), (b), (c), **split the time by the marks shown**, not evenly: a 3-mark part deserves a third of what an 8-mark part gets.